

TONIGHT

From a line to a prediction

Four weeks we described and tested. Tonight we predict: a number first, then a class, and then we see where one straight line stops being enough.

Data Analysis & Machine Learning for Business

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Two numbers differ. Is the difference luck?

EXAMPLE

Two groups of cats, two average weights. They are different. The numbers come from distributions, and comparing distributions by eye is hard and proves little.

step 1 Assume the null

The difference is luck: both groups come from one distribution. Alternative: group 2 is heavier.

step 2 Find evidence to reject it

This is where the tests differ.

THREE WAYS TO FIND THE EVIDENCE

permutation

Shuffle the labels

Mix the cats, split again, many times. How often does luck beat the real difference?

t-test

Compare the means

How big is the gap relative to the noise in each group?

Mann-Whitney

Compare the ranks

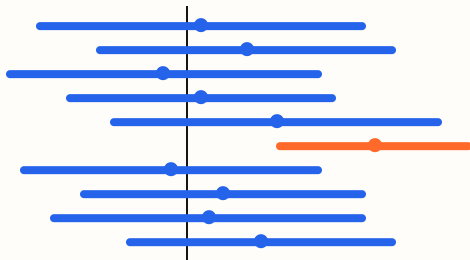
Who is heavier more often. Holds up when a long tail spoils the mean.

We reject the null or we don't. **We never prove the alternative.**

One measurement, and how far to trust it.

CONFIDENCE INTERVAL

We measured on a sample. The truth is unknown: the mean weight of **all** cats on the planet.



The truth doesn't move, the interval does. Build intervals this way forever and 95% cover the truth. You hold one and don't know which.

bigger sample → **narrower**

more confidence → **wider**

BOOTSTRAP

One week, productivity is **100**. How stable is that 100?

- 1 Draw a resample of the same size, **with replacement**
- 2 Compute the metric on it
- 3 Repeat 1,000 times, take the middle 95%

A formula covers only the mean, under assumptions. Median, share, p90 have none. **Bootstrap gets the width by simulation, not algebra.**

"How can we speed up picking?" From a number to a model.

CALCULATE

Count it

Picking speed across the warehouse

NOTICE

Numbers differ

Some places are slower than others

DELVE IN

See the shape

By rack inside a sector, by what sits in a cell

PROVE

Test it

Statistics: the difference is not luck

THEN: ML

Measure each factor

Effect of every feature, and what changes if we move it

WHAT DRIVES PICKING SPEED

- 1 **Location:** which rack inside the sector
- 2 **Cell structure:** how many items share the cell

So the answer is to **change placement**.

WHERE TO PLACE A NEW ITEM

A model: constraints plus a **target function** to minimize.
The two crucial terms:

Item homogeneous spread

Homogeneous cells: only this sneaker model

That case needed us to **predict, and know the effect of each factor**. Tonight we learn the tools.

Three steps, then a new homework.

01 A line through points

warehouse, minutes per task

02 One line between two classes

6,819 firms

03 When a line is not enough

trees, boosting, SHAP

04 The checklist, and HW2 goes out

due Oct 21

We don't know the formula, so we learn it from data.

EQUATIONS

distance = speed × time

The formula is known. Plug in numbers, get the answer.



MACHINE LEARNING

minutes = ? (picks, weight, ...)

The formula is unknown. Take a function with unknown parameters and **let the data estimate them.**

1. Train

DATA WE KNOW

X and y

features: picks, weight
target: minutes



FIT

Minimize the loss

loss: one number for how wrong the model is



RESULT

Parameters β

the learned numbers

2. Use

NEW ROW

X only

the target is unknown



PREDICT

Same formula, β

plug X into the learned formula



OUTPUT

$\hat{y}$

the guess of the target

The simplest model is a straight line.

task duration

 $\hat{y}$

guess of the target

=

minutes before
the first pick β_0

parameter

+

extra minutes
per pick β_1

parameter

×

number
of picks x

feature

FIT, THEN PREDICT

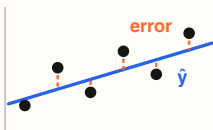
fit on 4,077 tasks $\rightarrow \beta_0 = 4.3, \beta_1 = 0.36$ predict a task of 30 picks $\rightarrow$ $\hat{y} = 4.3 + 0.36 \times 30 = \mathbf{15.1 \text{ min}}$

LOSS, AND HOW THE PARAMETERS ARE ESTIMATED

$$\text{error}_i = y_i - \hat{y}_i$$

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2$$

$$= \sum (y_i - \beta_0 - \beta_1 x_i)^2 \rightarrow \min$$



OLS (least squares) is just a method to find β_0 and β_1 : the pair that makes the loss smallest. `sm.OLS(y, X)`, always add_constant.

HOW ML WORK USUALLY GOES

- 1 **Find what drives the target.** Which features have the biggest influence?
- 2 **Build the best model** on that set of features.
- 3 **Judge it on test data,** rows the model has never seen.

Grade the model on rows it has never seen.

GOAL: EXPLAIN

How much does each factor matter?

Economists fit on **all the data** and read coefficients and p-values. No new rows to guess, so no split. Demo 1.

GOAL: PREDICT

How right will it be on new cases?

A model always looks good on the rows it learned from. **Out-of-sample test:** hide some rows during fit, grade only on them.

TRAIN_TEST_SPLIT, 6,819 FIRMS

test_size=0.3, stratify=y

train · fit here

4,773 firms · 154 bankrupt

test · 2,046 · 66

Stratify keeps the same 3.2% of bankrupt firms in both parts. A plain random cut could leave the test with too few of them.

Split **before everything else.**

Demo 01

One point per finished task, and the best straight line through 4,077 of them.

THE LINE OF THE EVENING

$$\text{minutes} = 4.3 + 0.36 \times \text{picks}$$

$R^2 = 0.40$. On average the line is off by 4.9 minutes.

- 1 The cloud: picks against minutes. **Draw the line by eye.**
- 2 Five toy tasks, then statsmodels along the schema
- 3 Real data: about 4 minutes of overhead, about 22 seconds per pick
- 4 The `.summary()` table: coefficient, std error, p-value, R^2
- 5 Add wrong scans: +0.26 min each, R^2 0.40 $\rightarrow$ 0.44
- 6 Two traps: scale and twins
- 7 Gauss-Markov on one card: error spread grows 3.8 $\rightarrow$ 8.1 min

The p-value comes back: **in a world where picks don't matter, how often is the slope this steep?**

A tiny coefficient can still matter. It depends on the units.

SAME WEIGHT, TWO UNITS

weight in kg **-0.04** min per kg

weight in grams **-0.00004** min per gram

SAME EFFECT

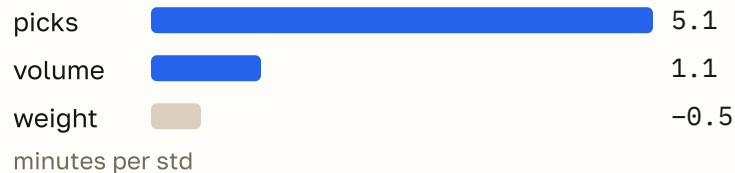
$$10 \text{ kg} \times -0.04 = -0.4 \text{ min}$$

$$10,000 \text{ g} \times -0.00004 = -0.4 \text{ min}$$

CURE: PUT EVERY FEATURE ON ONE RULER

$$z = (x - \text{mean}) / \text{std}$$

Now each coefficient means **minutes per one typical step** of that feature, and they can be compared.



A small coefficient does not mean an **unimportant feature**.

Twin features steal each other's credit.

THE TWINS

picks ↔ **weight**

More picks, heavier task. They move together.

0.61

correlation

COEFFICIENT OF WEIGHT, IN TWO MODELS

A: weight alone **+0.27** min per kg, heavier takes longer

B: picks + weight **-0.04** heavier is faster?!

In B, **picks already explains the minutes**. Weight gets only what is left, and its sign flips. Add a near-copy of picks and both lose significance: $p = 0.28$ and 0.81 .

Cure: check correlations before the model, and keep one twin.

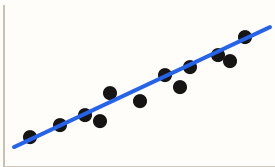
Ask Why would heavier tasks be faster?

Two kinds of prediction: a number or a class.

REGRESSION

How much? How long?

The line passes **through** the points.



TARGET Y 12.4 min, 7.1 min, 20.3 min ...

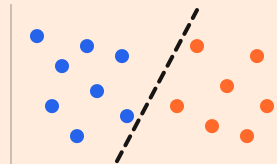
OUTPUT a number

EXAMPLES task duration, demand, price

CLASSIFICATION

Will it happen? Yes or no.

The line goes **between** two groups.



TARGET Y 0 healthy, 1 bankrupt

OUTPUT a probability, then a class

EXAMPLES default, churn, fraud

6,819

listed firms, Taiwan, 1999 to 2009

95

financial ratios

3.2%

went bankrupt

UCI 572

Liang et al., 2016

Why not the line from Demo 1?

THE FIRST TRY

$$\hat{y} = \beta_0 + \beta_1 \times \text{debt}$$

bankrupt = 1, healthy = 0.

Read $\hat{y}$ as the probability.

WHAT IT SAYS

25%

of test firms get a **negative** probability.

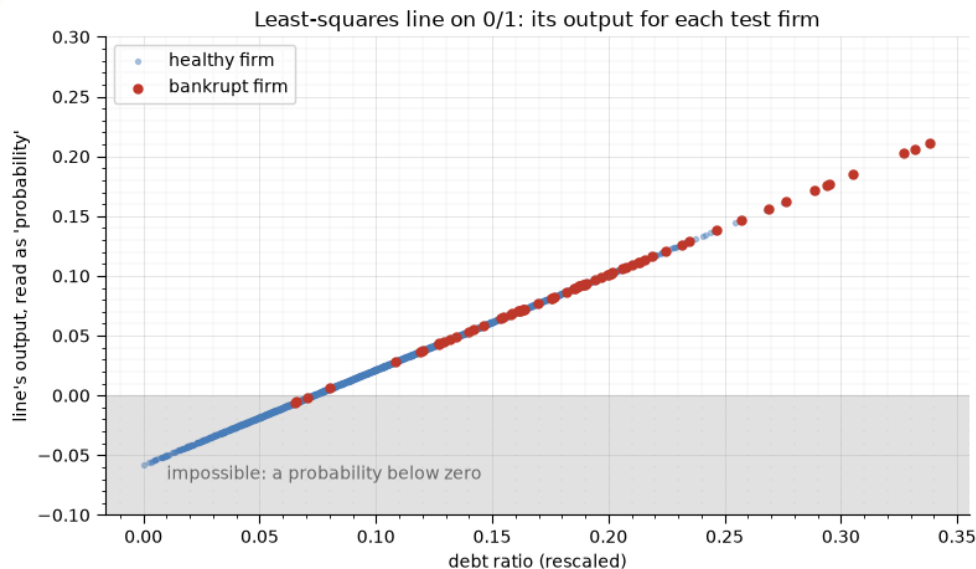
Lowest: -0.058.

a line

$-\infty \dots +\infty$

a probability

[0, 1]



A line is unbounded. A probability must stay **between 0 and 1**.

Logistic regression is a line pushed through a sigmoid.

step 1 $z = \beta_0 + \beta_1 \times \text{debt}$

the same weighted sum, a score from $-\infty$ to $+\infty$

step 2 $p = 1 / (1 + e^{-z})$

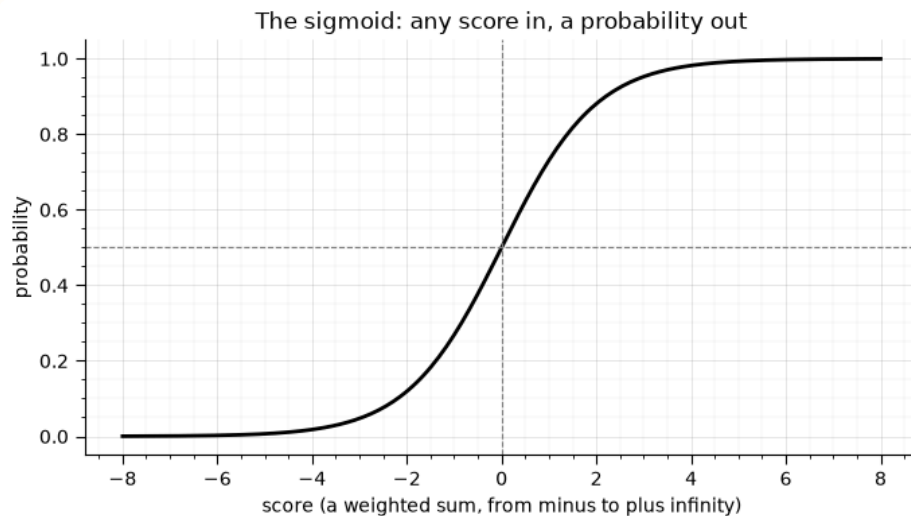
the sigmoid: any score in, a probability out

$z = -6$
 ≈ 0

$z = 0$
0.5

$z = +6$
 ≈ 1

Parameters by **maximum likelihood**, the classification cousin of least squares. sm.Logit instead of sm.OLS.



A threshold turns a probability into a class.

THE RULE

$p \geq 0.5 \rightarrow$ bankrupt

On the debt axis: cut-off at 0.282.

DEBT RATIO IN, RISK OUT

a firm's debt ratio

0.11 median

0.20

0.30 liabilities = assets

These are firms' values, not coefficients. The sigmoid turns each value into a risk.

P(bankrupt)

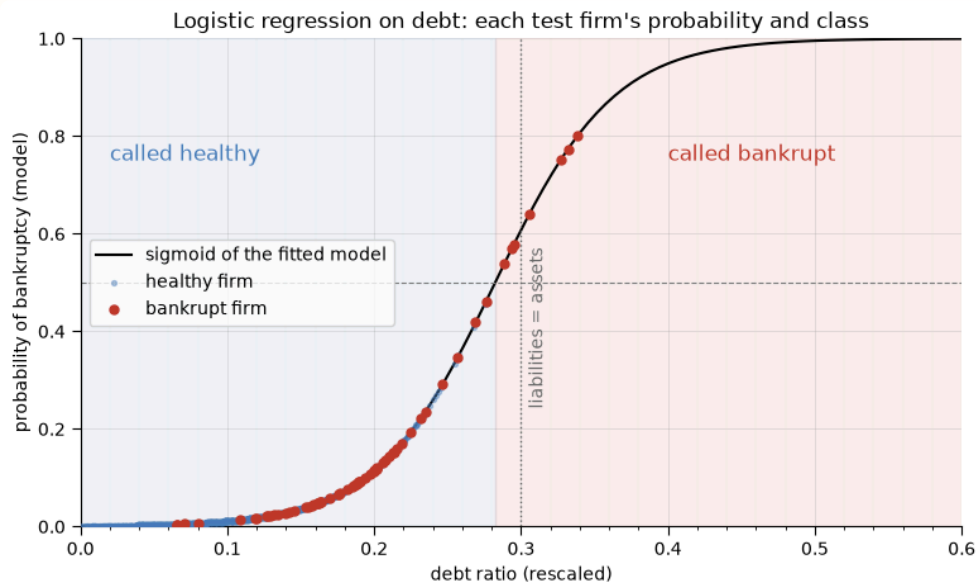
1.4%

12%

61%

7 of 66 bankruptcies caught

Red dots on the blue side were missed. Debt alone is not enough.



0.5 is a default, not a law. Lower it to catch more, at the price of false alarms. Session 6.

Demo 02

Financial ratios of real companies. For once, you are the experts.

THE QUESTION

Can a few financial ratios tell bankrupt firms from healthy ones?

Altman answered yes in 1968: five ratios and one line between the two groups.

- 1 Find the source page before you trust a column
 - 2 Clean the names, drop Net Income Flag: it contradicts its own description
 - 3 Ratios are rescaled: read directions, not units
 - 4 A toy: two clouds and one line between them
- ↓ Then: the split, the line on 0/1, the sigmoid, the threshold

Altman's five ratios, refitted on our firms. One sign is wrong.

ALTMAN'S RATIO	EXPECT	WEIGHT	P
Working capital / assets	–	–0.25	0.03
Retained earnings / assets	–	+0.18	0.01
EBIT / assets (ROA)	–	–1.16	<0.001
Equity / liabilities (debt ratio)	+	+1.09	<0.001
Sales / assets (turnover)	–	–0.63	<0.001

TWINS AGAIN

Retained earnings and ROA move together, corr 0.63. Same trap as Demo 1, new data.

IN REAL BANKS

Not Altman's five. Banks under Bank of Russia **Regulation 483-P** build their own default models on their own data and ratios. The method is often the same **logistic regression**: each coefficient can be explained and checked.

Altman's Z grows with safety. Ours predicts bankruptcy, so **protective ratios get negative weights**.

A straight-A student that almost never raises its hand.

"Everyone is fine"

no model at all

ACCURACY

96.8%

0 of 66 caught

Logistic regression

as it comes

ACCURACY

96.9%

12 of 66 caught

With class weights

bankrupt firms weigh as much, together,
as healthy ones

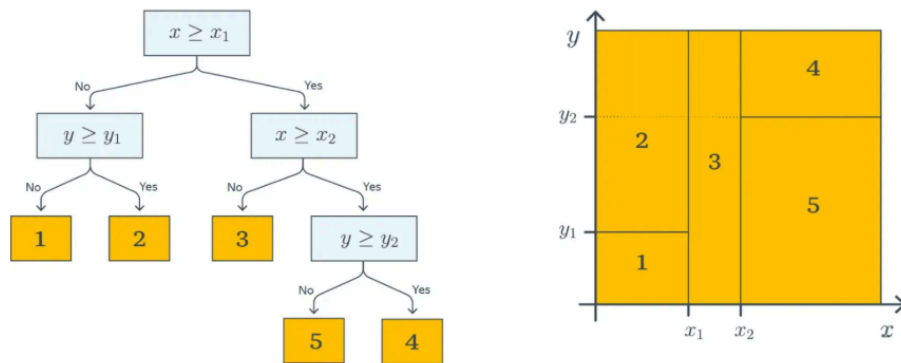
ACCURACY

86.3%

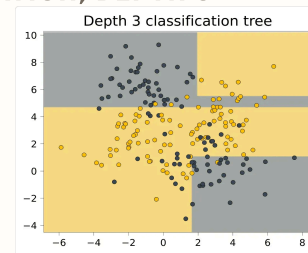
57 of 66 caught, **272** false alarms

Which is better depends on **the price of each mistake**. Session 6.

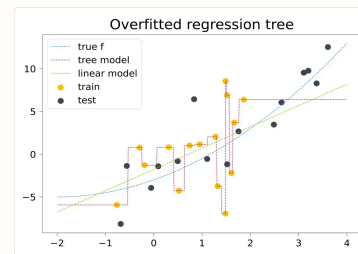
A tree asks yes/no questions and cuts the space into boxes.



CLASSIFICATION, DEPTH 3



REGRESSION: A STAIRCASE



- 1 Each question: one feature, one threshold. The leaf holds the answer.
- 2 Class in each box, or one number: the average of its rows.
- 3 Too deep, and it memorises the training rows. **Limit depth, trust the test.**

Demo 03

Trees and boosting, on the warehouse and on the firms.

SHELF LEVELS, SECONDS PER PICK

A line says **+0.44 s per level**, significant. The averages form a U: floor 9.3, middle 7.6, top 11.

Significant is not the same as right.

- 1 Shelf levels: the line misses the U, a tree catches it
- 2 CatBoost on the firms, with class weights: one prompt, a dozen lines
- 3 Train firms: 154 of 154 caught. **It has memorised them**
- 4 Instead of coefficients: importance, and SHAP for one firm

The train number is **not a grade**.

Two models, and no winner yet.

THE BASELINE

Weighted logit

57

caught

272

false alarms

THE CHALLENGER

CatBoost, weighted

34

caught

accuracy 96.4%, below "everyone is fine"

41

false alarms

INSTEAD OF COEFFICIENTS

Top importance

- 1 interest rate on debt
- 2 Quick Ratio
- 3 borrowing dependency

Same families as Altman, different representatives. **SHAP** splits one firm's verdict among its ratios.

Who wins depends on **the price of each mistake**. That is session 6.

Six lines, each one earned tonight.

01 A line first. It is readable, and it is your baseline.
The weighted logit.

02 Read coefficients only after standardizing.
Kilograms and grams.

03 Check twins before trusting a sign. Weight next to
picks.

04 Significant is not the same as right: look at the
shape. Shelf levels.

05 With 97% of one class, accuracy flatters. Count
what you catch and what you miss.

06 The train number is not a grade. 154 of 154.

The assistant writes the code in one prompt. **Every line on this list is what you do around it.**

One real dataset, one yes-or-no question.

One of five, by DM, about three of you per dataset:

credit cards

bank marketing

home-equity loans

Lending Club

hotel bookings

Part 0 Clean. Watch for columns known only after the outcome.

Part 1 Explore with charts: what goes with “yes”? End with hypotheses.

Part 2 The imbalance fork: choose how to treat the rare class.

Part 3 A readable logistic regression. Test your hypotheses.

Part 4 Can boosting beat it? Worth losing a readable model?

Part 5 The cheapest threshold. Each dataset prices mistakes. [session 6](#)

Part 6 What drives it: SHAP, two cases a manager accepts.

Part 7 A memo, ten sentences, for the person who decides.

YOUR DATA IS NOT TAIWAN

Text columns and gaps. Read “**Before you start**” in the README first.

GRADING

8 a complete, correct job

+2 one extra, done well

AI is allowed. You must be able to explain every line.

NEXT WEEK, FIRST THING

The quiz. 25 minutes, closed book, understanding, not memory.

Parts 0 to 4 **tonight** | Due **Oct 21, 18:00**, notebook and html by DM

Five ideas to keep.

- 01 Regression is the best line through points. Read coefficients, p-values and R^2 .
- 02 Scale and twins are the two classic traps.
- 03 Classification is one line between classes. Logistic regression is a line pushed through a sigmoid.
- 04 At 3% positives accuracy flatters. Count catches and misses.
- 05 When a line isn't enough: trees, boosting, and SHAP instead of coefficients.

NEXT WEDNESDAY

Tonight two models each won in a different way, and we couldn't pick. Next week we put a price on each mistake and let the money decide. **The quiz comes first.**